PDB: healthy and variant structure

This document selects and characterizes the wild type and crystal structures of human PMM2 (UniProt O15305) that will be used for the 3D comparison and summarizes what they show about the role of the variant.

RCSB PDB holds seven structures of human PMM2. Five come from the same study (Briso-Montiano 2022) 7O0C, 7O1B, 7O4G, 7O5Z and 7O58 and two, 2Q4R and 2AMY are older entries from other groups.

We choose 7O1B (healthy) and 7O58 with mutation (T237M).

We have chosen them because according to the original study, both crystals were co-crystallized with the activator glucose-1,6- bisphosphate (G16), so they were prepared in the same way and differ in the variant. We initially considered 7O4G, the wildtype structure with the best resolution (2.66 Å), but it was obtained by soaking pre-formed crystals with the activator, so a comparison with 7O58 would mix the effect of the mutation with a different preparation.

Comparison:

- **PDB data 7O1B**

TITLE, ID AND KEY INFO:
Human phosphomannomutase 2 (PMM2) wild-type co-crystallized with the activator glucose 1,6-bisphosphate

Experimental Data Snapshot:

Method: X-RAY DIFFRACTION
Resolution: 3.08 Å
R-Value Free:
0.257 (Depositor), 0.256 (DCC)
R-Value Work:
0.236 (Depositor), 0.232 (DCC)

Primary citation:
Briso - Montiano and collaborators (2022) J Inherit Metab Dis **45**: 318-333

Macromolecules description:

Molecule	Phosphomannomutase 2
Chains	A, B
Sequence length	248
Organism	*Homo sapiens*
Mutation(s)	0
Gene name	PMM2
EC number	5.4.2.8
UniProt	O15305

Small molecules / Ligand types:

Ligand ID	Name	Formula	Chain(s) (author numbering)	Role
G16	1,6-di-O-phospho-alpha-D-glucopyranose (glucose-1,6-bisphosphate)	$C_6H_{13}O_{12}P_2$	A	Enzyme activator
CL	Chloride ion	Cl	B	Ion from crystallization conditions
MG	Magnesium ion	Mg	A (two ions), B (two ions)	Metal ion, a likely cofactor

- **PDB data 7O58:**

TITLE, ID AND KEY INFO:
Human phosphomannomutase 2 (PMM2) with mutation T237M in complex with the activator glucose 1,6-bisphosphate

Experimental Data Snapshot:
Method: X-RAY DIFFRACTION
Resolution: 1.97 Å
R-Value Free:
0.230 (Depositor), 0.229 (DCC)
R-Value Work:
0.183 (Depositor), 0.182 (DCC)

Primary citation:
Briso - Montiano and collaborators (2022) J Inherit Metab Dis **45**: 318-333

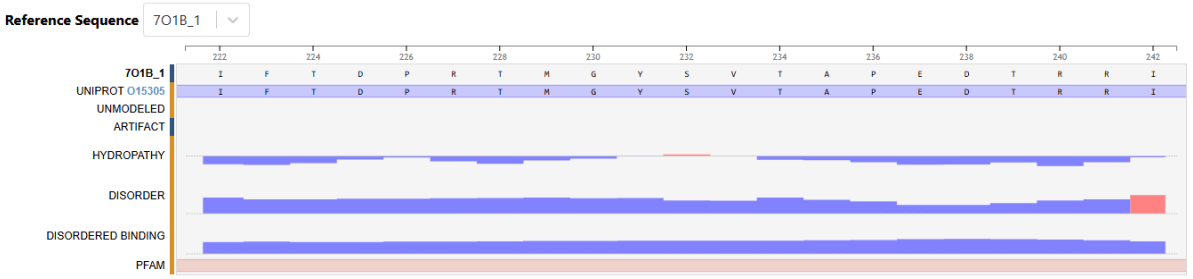
Macromolecules description:

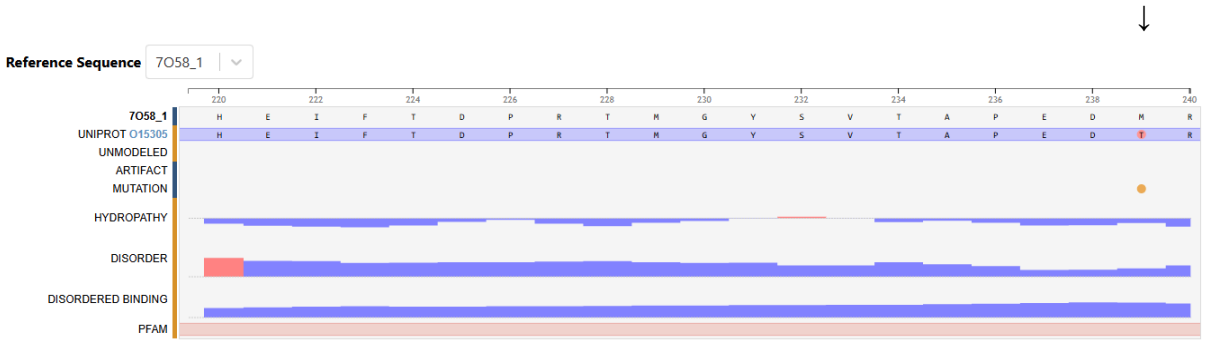
Feature	**Value**
Molecule	Phosphomannomutase 2
Chains	A, B
Sequence length (amino acids)	248
Organism	*Homo sapiens*
Mutation(s)	1
Gene name	PMM2
EC number	5.4.2.8
UniProt accession	O15305

Small molecules:

Ligand ID	Name	Formula	Chain(s) (author numbering)	Role
G16	1,6-di-O-phospho-alpha-D-glucopyranose (glucose-1,6-bisphosphate)	$C_6H_{13}O_{12}P_2$	A	Enzyme activator
GOL	Glycerol	$C_3H_8O_3$	A (at least five copies)	Crystallization additive
CL	Chloride ion	Cl	A	Ion from crystallization conditions
MG	Magnesium ion	Mg	A (two ions), B (two ions)	Metal ion, a likely cofactor
NA	Sodium ion	Na	A (two ions), B (one ion)	Ion from crystallization conditions

Numbering in the PDB sequence viewer is to positions ahead of UniProt, because the construct has 248 residos versus 246. UniProt Thr237 therefore corresponds to position 239 in the viewer. In both chains of 7O1B this residue is threonine and is modelled and in both chains of 7O58 it is methionine and is modelled. In RCSB, the position falls within a helical segment, consistent with the α9 helix described by the authors (Briso- Montiano 2022). Only an isolated residue in chain B of 7O1B (around viewer position 204), far from residue 237, is not modelled.





Biological assembly:

Both structures are homodimers (Global Stoichiometry: Homo 2-mer, A2; Global Symmetry: cyclic, C2). This agrees with the dimeric form of PMM2, which is considered essential for the activity (Briso-Montiano 2022, Hay Mele 2025). Therefore, 7O1B and 7O58 can be compared dimer against dimer.

Ligands:

Both structures contain the activator G16, magnesium and chloride ions. 7O58 additionally contains glycerol and sodium ions, which derive from the crystallization conditions. Magnesium is a likely cofactor, and Thr237 interacts with Thr16 in the loop that coordinates one magnesium ion (Briso-Montiano 2022).

Model quality:

7O1B (3.08 Å) is of medium resolution and 7O58 (1.97 Å) of high resolution. R - work is 0.236 for 7O1B and 0.183 for 7O58; higher value for 7O1B is consistent with its lower resolution, on both indicate acceptable agreement with the experimental data. 7O1B has a clashscore of 7, 0,2% of residues in disallowed Ramachandran regions and no unusual side-chain conformations.

Conclusion:

We selected 701B and 7058 because both contain the activator and come from the same study. The T237M structure is nearly identical to the wild type one, so the pathogenic effect is explained by lower stability, probably through the loss of a hydrogen bond with Thr16, and not by a major structural change.